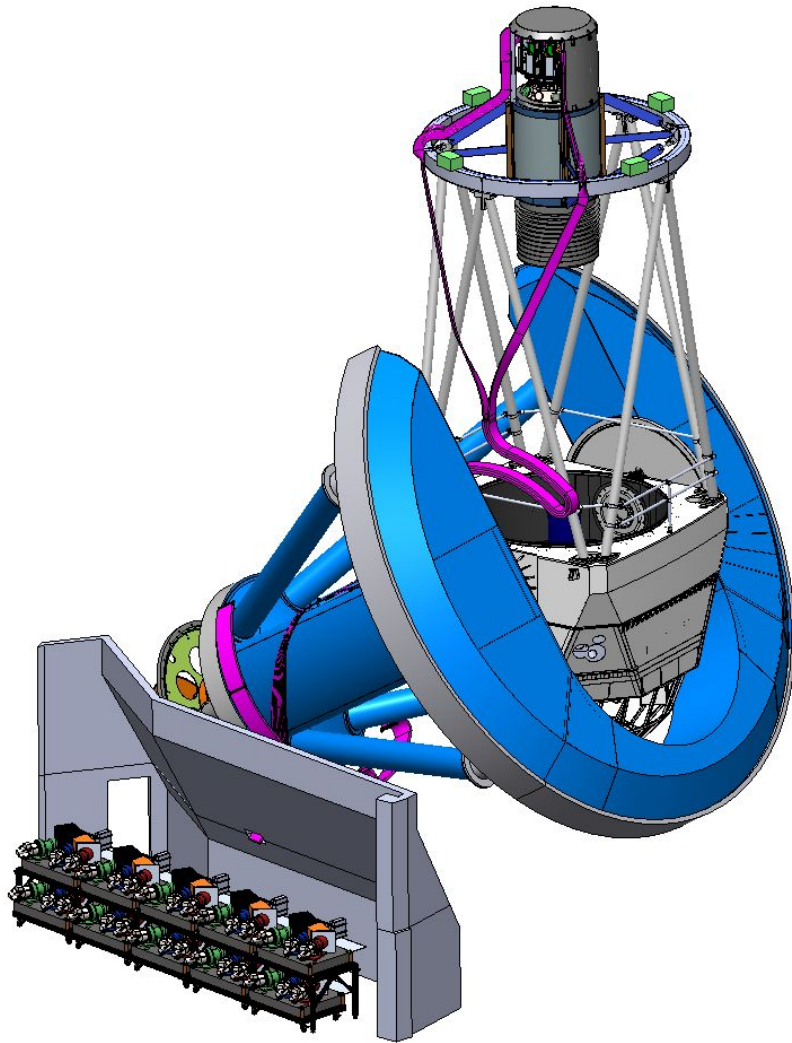


Observing at High Airmass

David Kirkby, UC Irvine

[DESI-8586](#)



Dark Energy Spectroscopic Instrument
U.S. Department of Energy Office of Science
Lawrence Berkeley National Laboratory



Overview

Identify observations of tiles at high airmass.

Look at motion of sky relative to focal plane during the exposure due to changing airmass.

Compare ideal guiding with actual guider performance.

Look at possible observing efficiency improvements:

- guiding on rotation
- shorter exposures



Dataset

10 exposures of 7 tile designs at airmass > 2.0 with > 600s exptime:

	expid	night	tileid	airmass	skyra	skydec	mjd_obs	exptime	etcteff	etcsplit	parallactic	mountel	mountha	acqfwhm	adc1	adc2	fieldrot
0	260708	20241101	83482	2.950	15.003050	-38.02051	60616.214202	1801.9	344.7	1	-6.766443	19.65	-7.680	1.596	264.0	82.7	0.041371
1	266909	20241205	83482	2.918	15.001742	-38.02060	60650.130106	1803.3	127.8	1	-3.930205	19.89	-4.516	1.559	266.9	85.6	0.041371
2	265751	20241128	83514	2.908	60.000950	-38.02063	60643.278510	1511.7	180.8	1	-2.425841	19.97	-2.857	1.370	268.6	87.3	0.137451
3	266760	20241204	83482	2.905	15.000808	-38.02063	60649.139700	1803.0	339.2	1	-1.697374	19.99	-2.039	1.468	268.9	87.6	0.041371
4	260432	20241030	83481	2.377	15.002342	-33.02055	60614.223690	1809.0	589.4	1	-5.659690	24.75	-6.218	1.218	265.3	83.9	0.038883
5	262660	20241113	83501	2.368	15.001571	-33.02060	60628.189133	1800.1	76.7	1	-4.424532	24.85	-4.895	1.113	267.0	85.7	0.038883
6	260566	20241031	83481	2.357	14.998979	-33.02063	60615.243427	1284.9	331.9	1	1.924042	24.99	1.889	1.370	273.1	91.8	0.038883
7	254592	20240922	83024	2.274	35.715929	-5.76324	60576.250898	1327.3	80.2	1	-50.125868	25.86	-54.572	1.432	220.6	39.2	0.073552
8	261833	20241107	83500	2.012	15.002200	-28.01966	60622.200830	1800.1	45.5	1	-6.275484	29.69	-6.586	2.507	281.4	67.3	0.036942
9	259630	20241025	83480	2.006	15.001892	-28.01965	60609.239752	998.7	514.5	1	-5.072419	29.80	-5.357	1.137	282.5	67.8	0.036942
10	259631	20241025	83480	1.994	15.001892	-28.01965	60609.252702	908.8	489.1	2	-0.504474	30.01	-0.767	1.137	288.4	72.2	0.036942
11	259497	20241024	83479	1.767	15.002821	-23.01700	60608.232499	1730.6	1005.3	1	-9.039882	34.36	-8.959	1.053	295.0	47.7	0.035443
12	261346	20241104	83499	1.760	15.002208	-23.01695	60619.206076	1111.1	181.1	1	-7.716549	34.53	-7.650	1.963	297.5	48.9	0.035443
13	208022	20231204	83024	1.477	35.692242	-5.76047	60283.288070	782.8	64.0	1	35.444116	42.60	30.133	0.879	357.3	75.4	0.073552



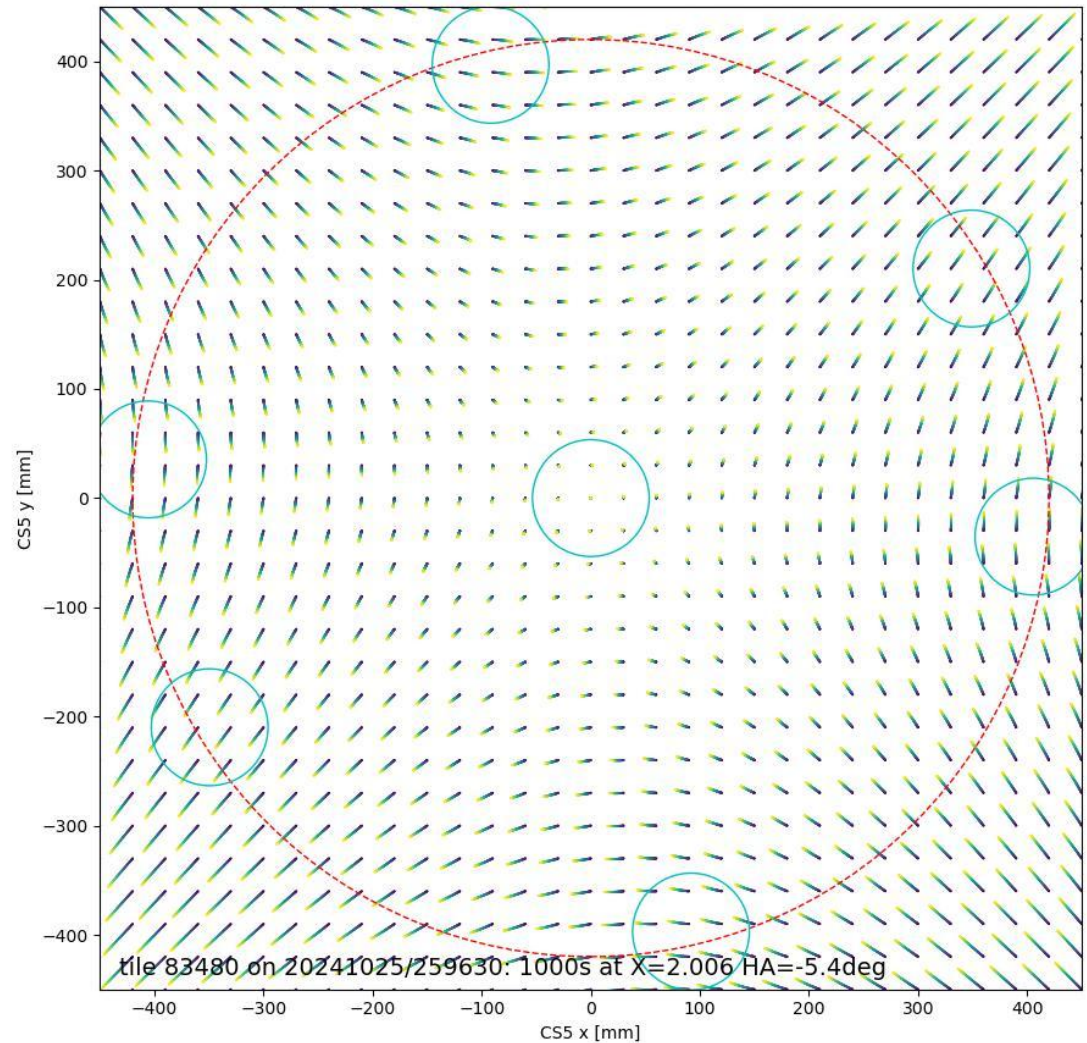
Ideal Guiding at $X=2.0$

Trails show sky motion relative to focalplane during 1000s exposure
magnified 1000x

Ideal guiding in (x,y) pins boresight at center and removes any dipole.

Remaining quadrupole is irreducible.

Smaller circles show $107\mu\text{m}$ fiber at each GUIDE location.

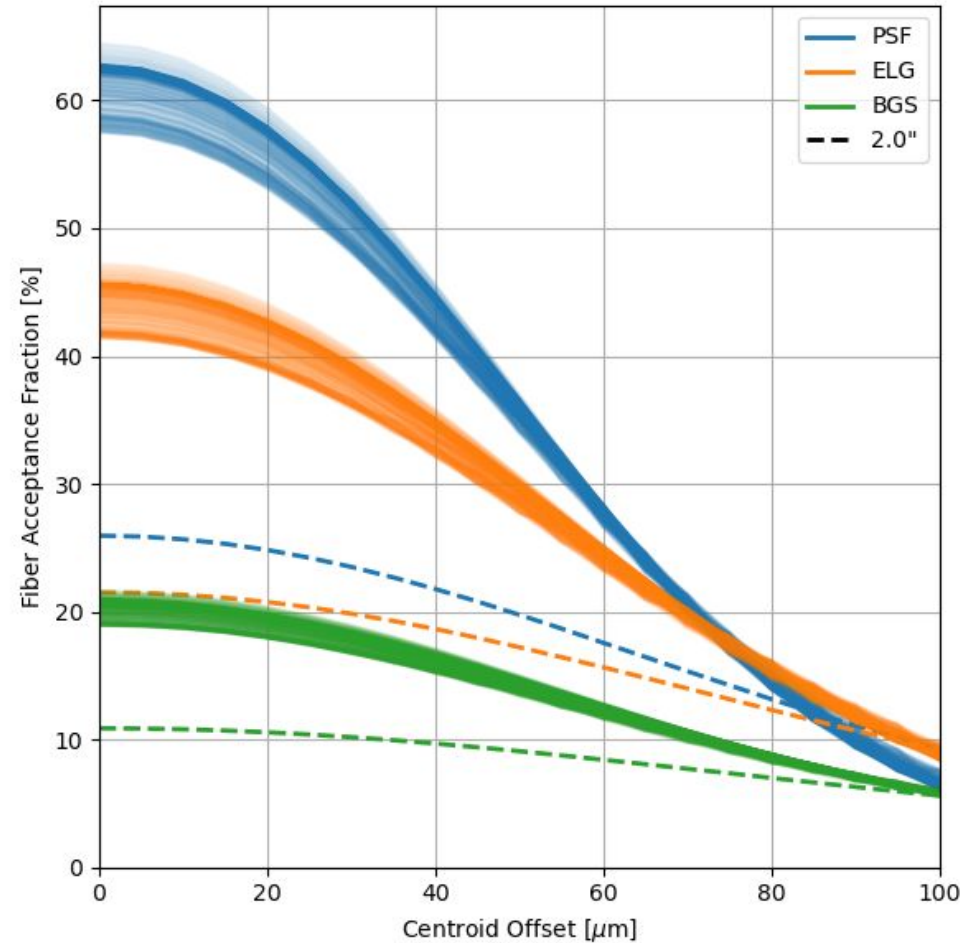
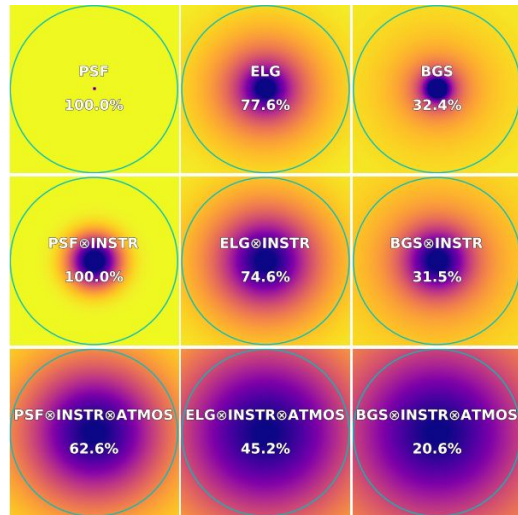


Ideal Guiding at $X=2.0$

In this example, sky moves by up to $15\mu\text{m}$ over 1000s.

How does that impact throughput?

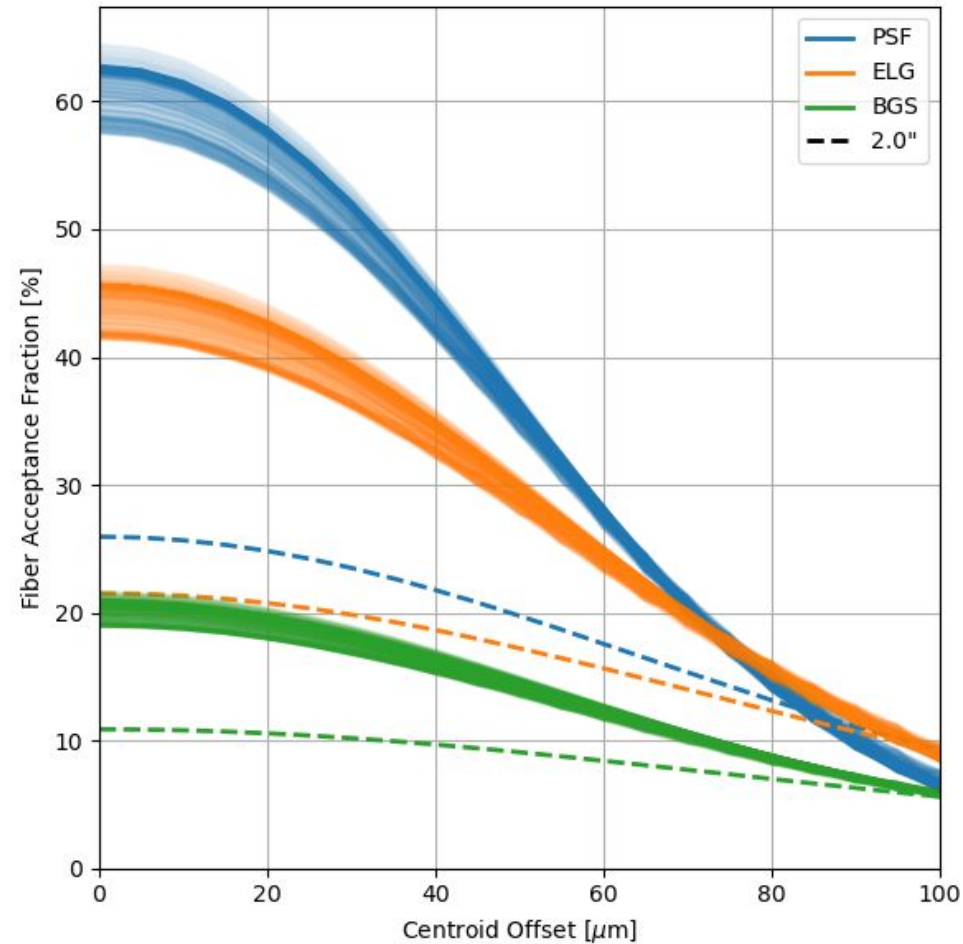
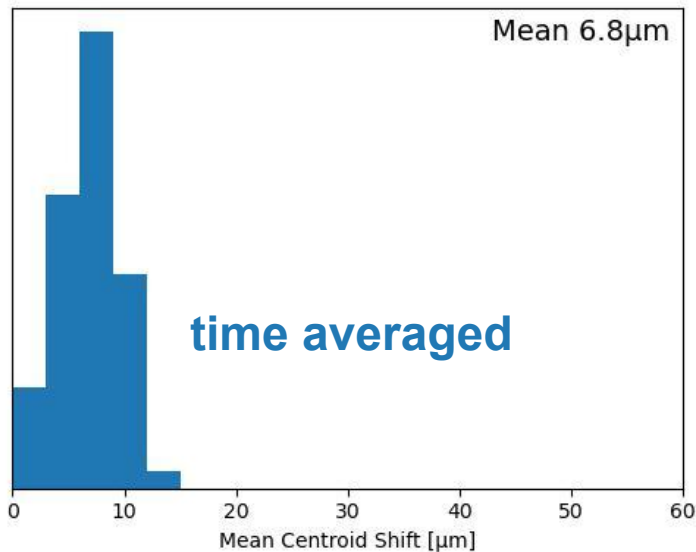
Answer depends on seeing, target size, focal-plane position:



Ideal Guiding at $X=2.0$

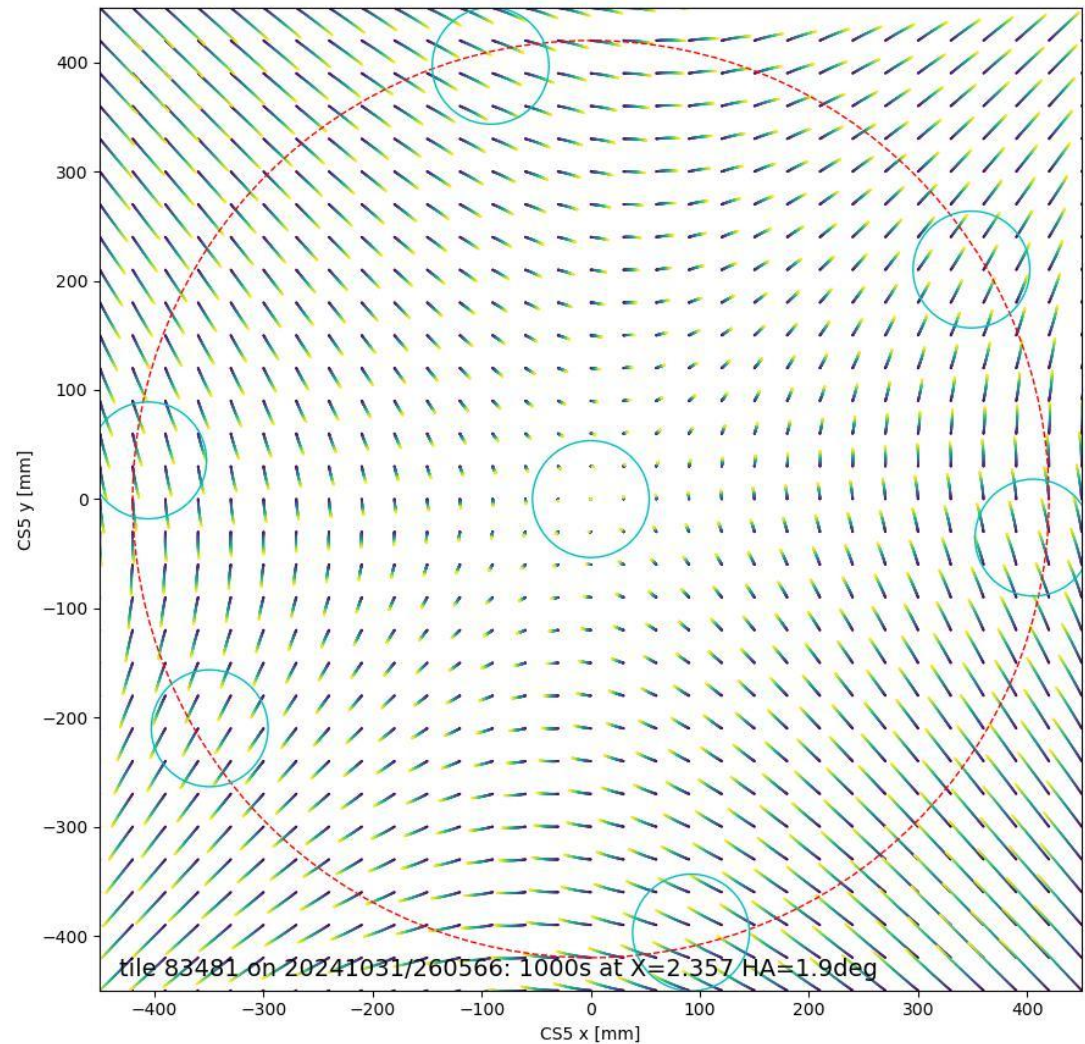
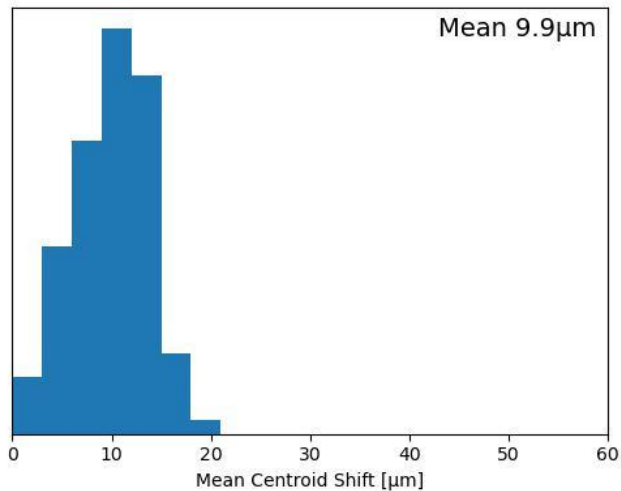
Relevant quantity is the mean centroid shift averaged over:

- the exposure duration
- focal plane position



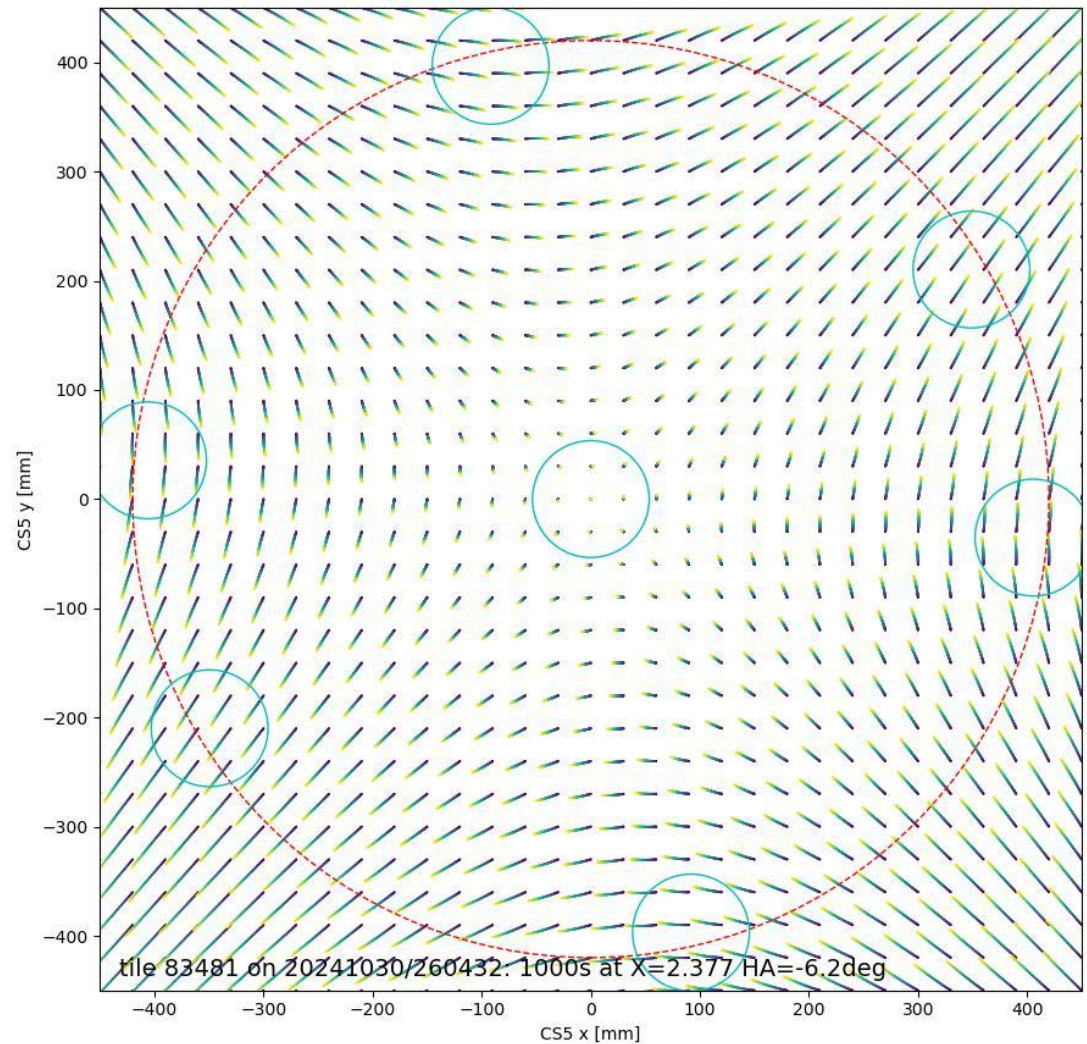
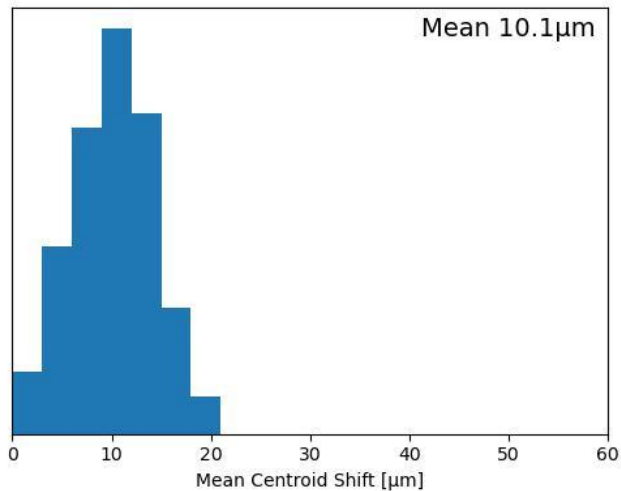
Ideal Guiding at $X=2.4$

1000s observation at initial
 $X = 2.36$, $HA=+1.9^\circ$



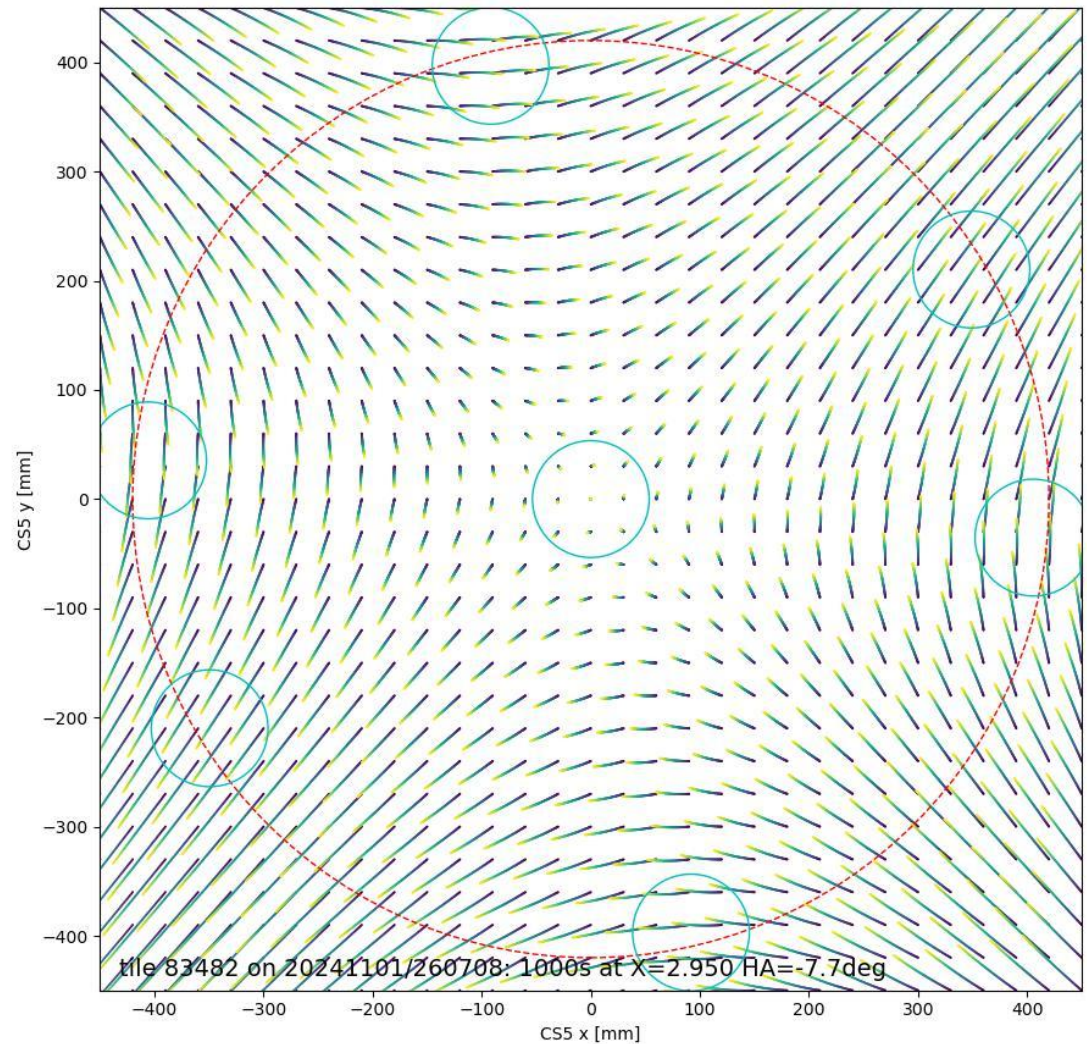
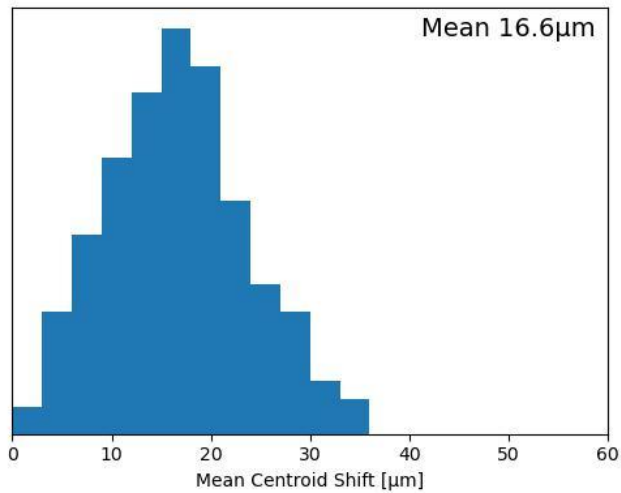
Ideal Guiding at X=2.4

1000s observation at initial
 $X = 2.38$, $HA = -6.2^\circ$



Ideal Guiding at X=3.0

1000s observation at initial
 $X = 2.95$, $HA = -9.7^\circ$



Guide Star Analysis

Track the motion of the patch of sky seen by each GFA.

Use ETC results rather than guider corrections:

- stored in ETC json file as thru/dx_gfa, thru/dy_gfa
- in units of GFA pixels
- one number per GFA (not per star)

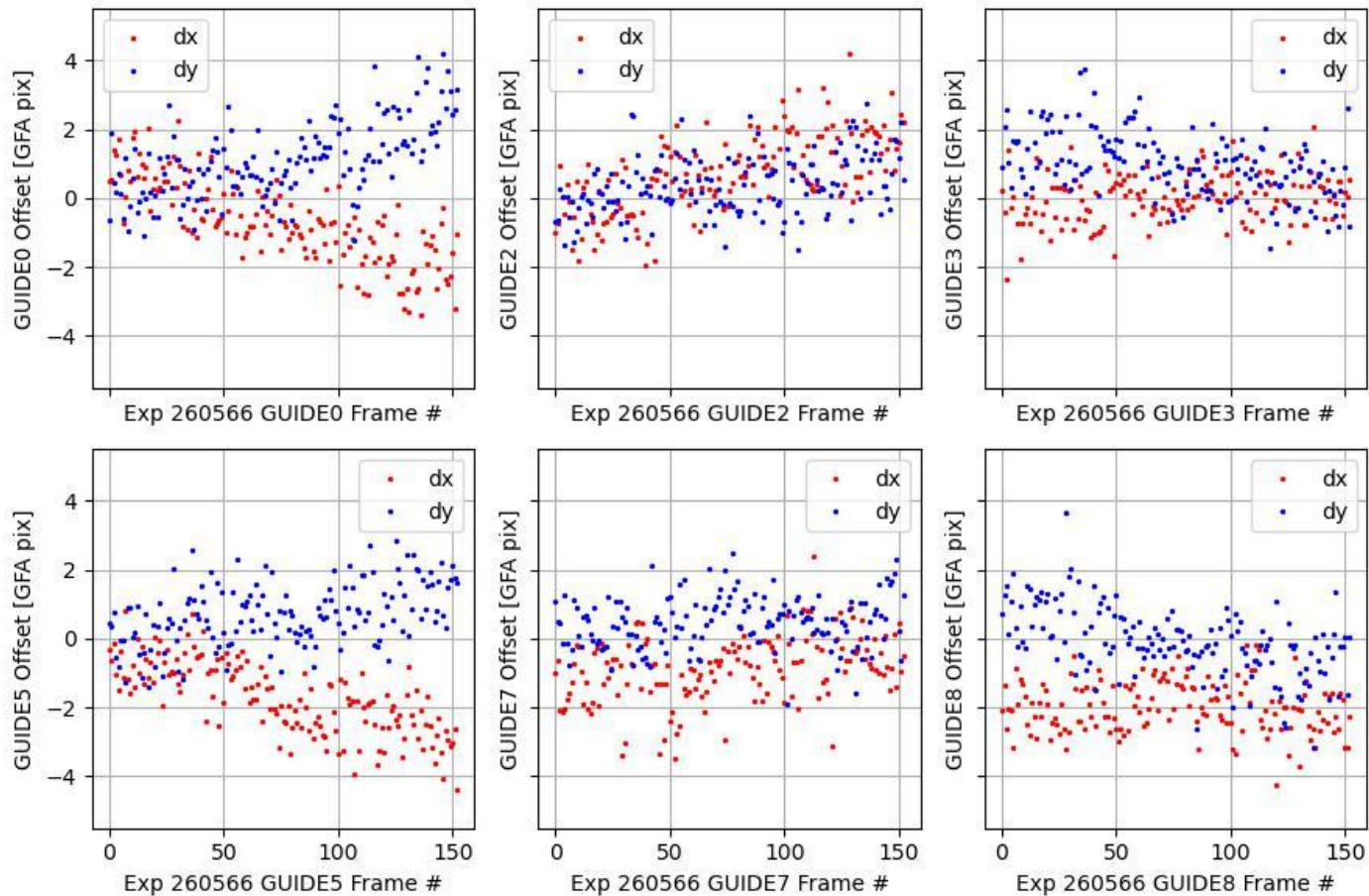
Fit x,y trends in each GFA with smoothing.

Convert from GFA pixels to CS5 to see global pattern.



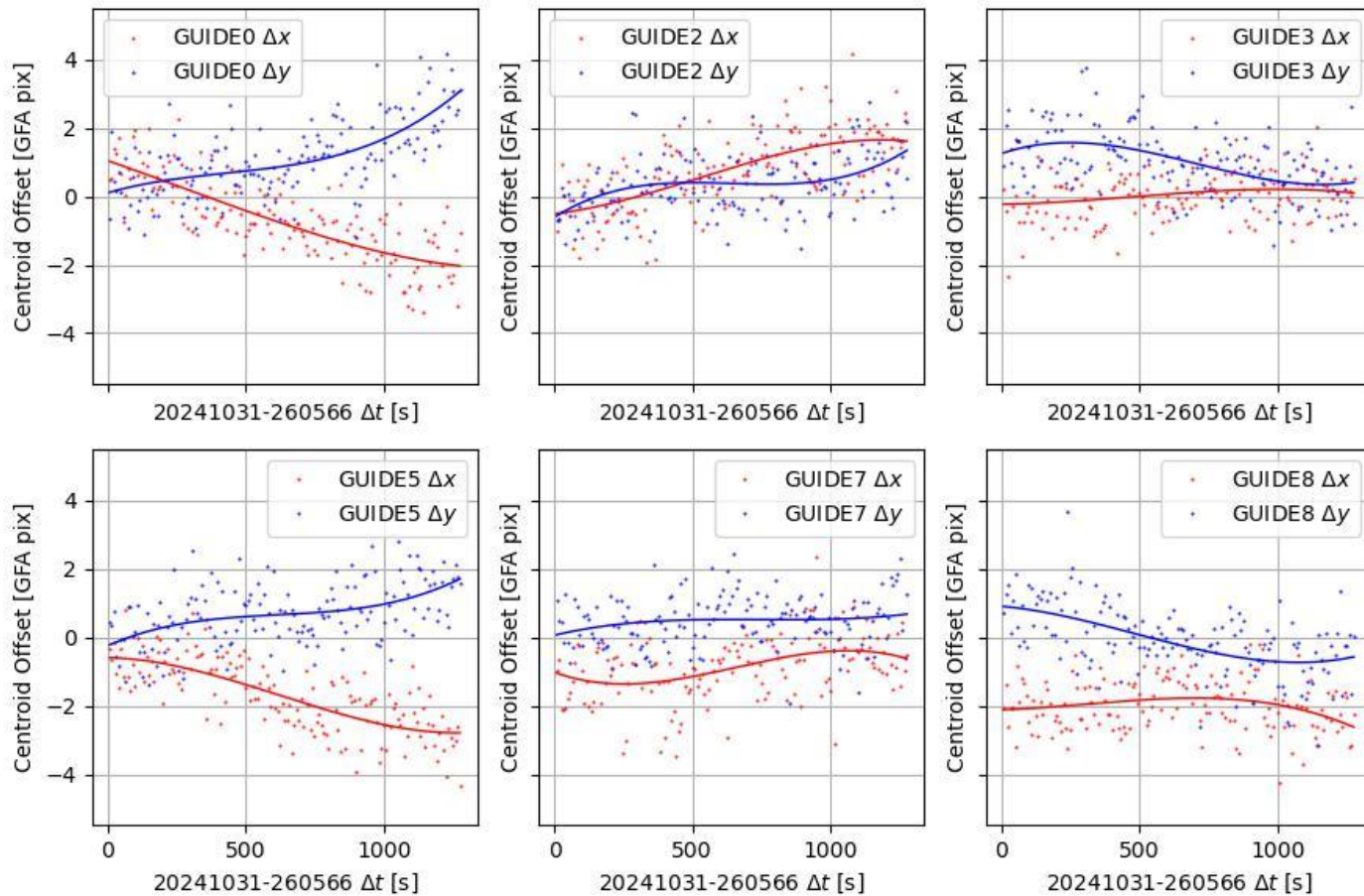
Guide Star Offsets in EXPID 260566 at X=2.3

Calculate centroid drift of guide stars on each GUIDE camera:



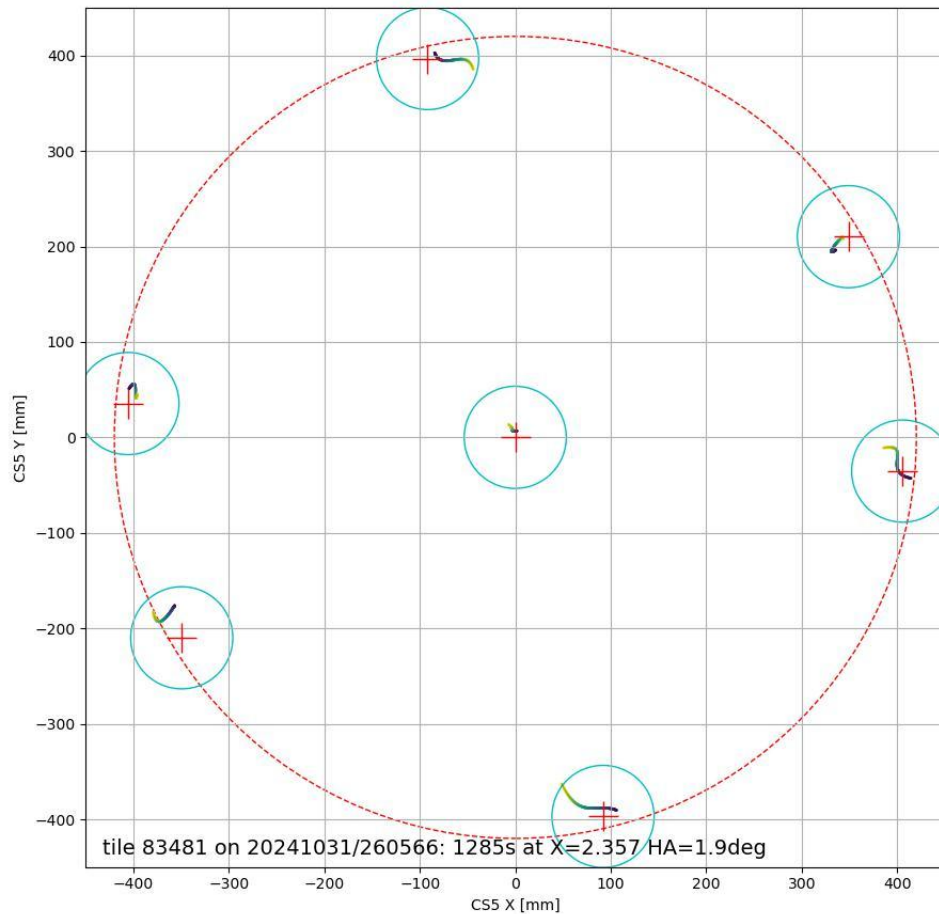
Guide Star Offsets in EXPID 260566 at X=2.3

Fit trends with smoothing:



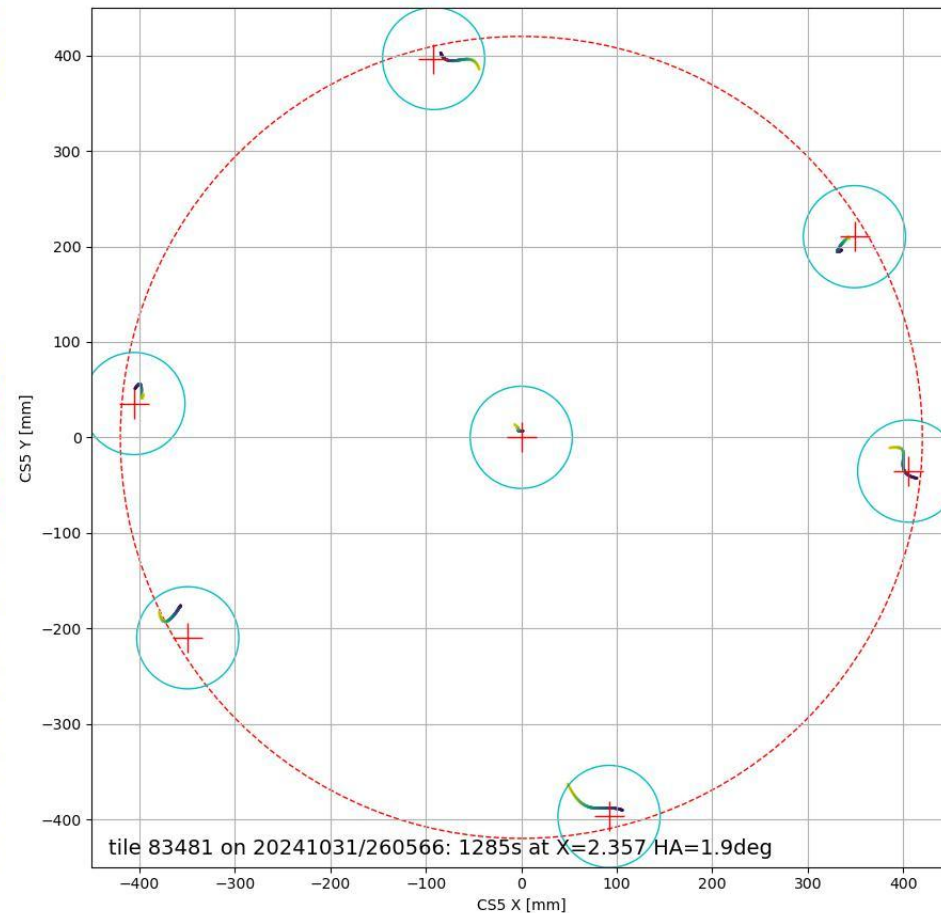
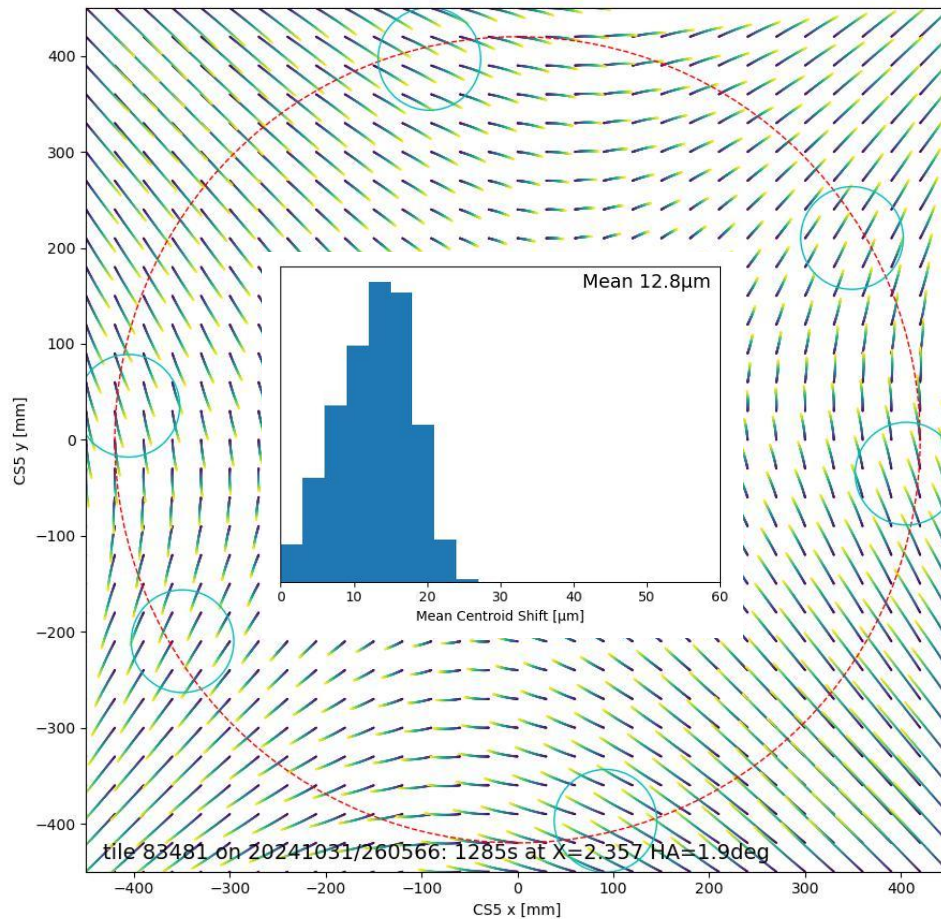
Guide Star Offsets in EXPID 260566 at X=2.3

Convert from GFA pixels to CS5 and combine:



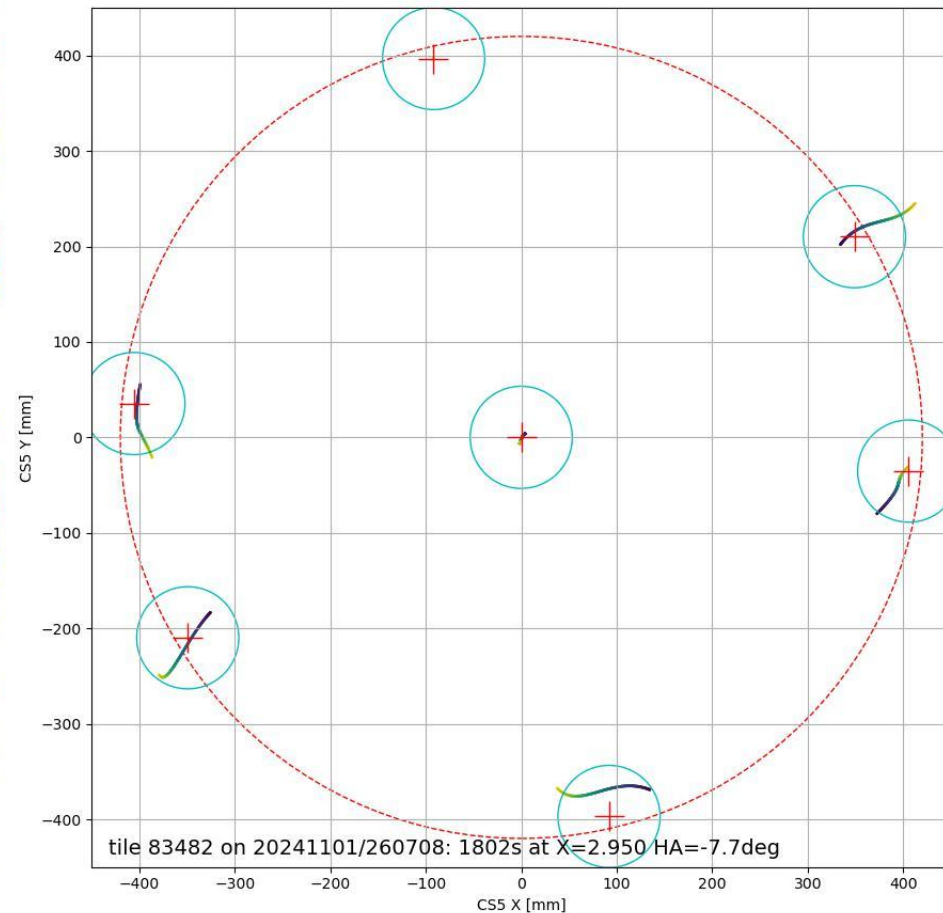
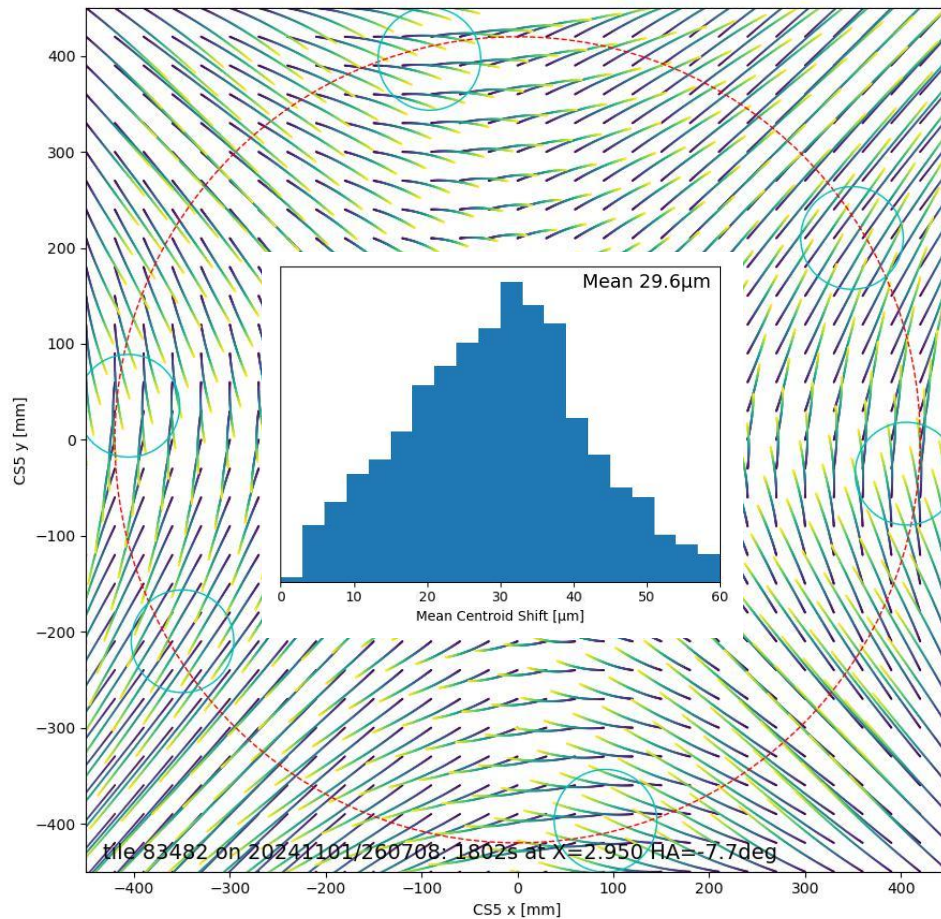
Guide Star Offsets in EXPID 260566 at X=2.3

Compare with ideal guiding for actual exposure time (1285s):



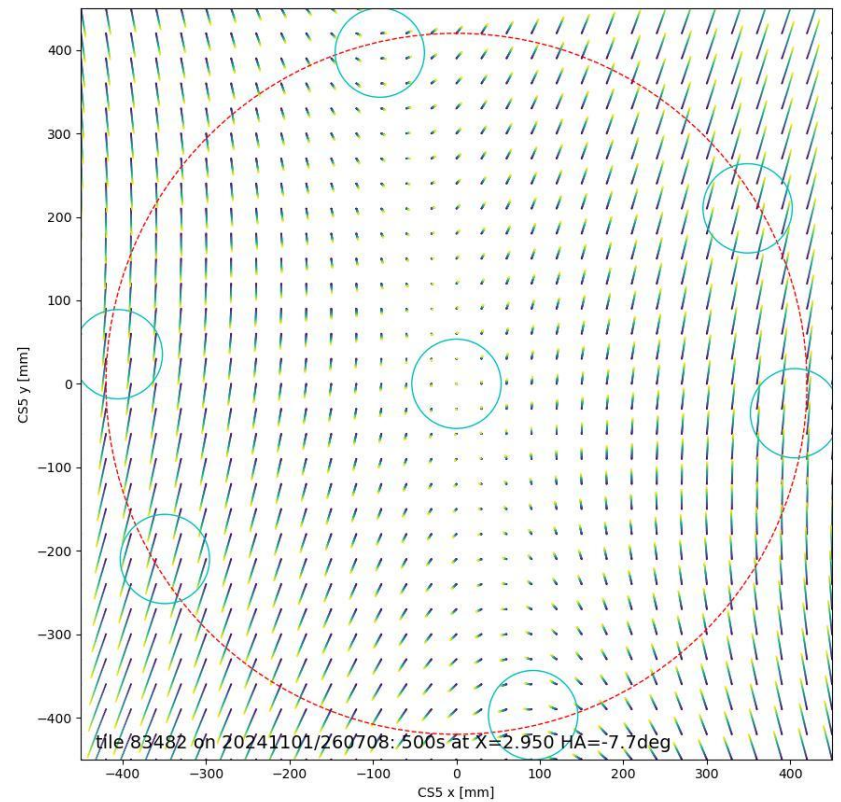
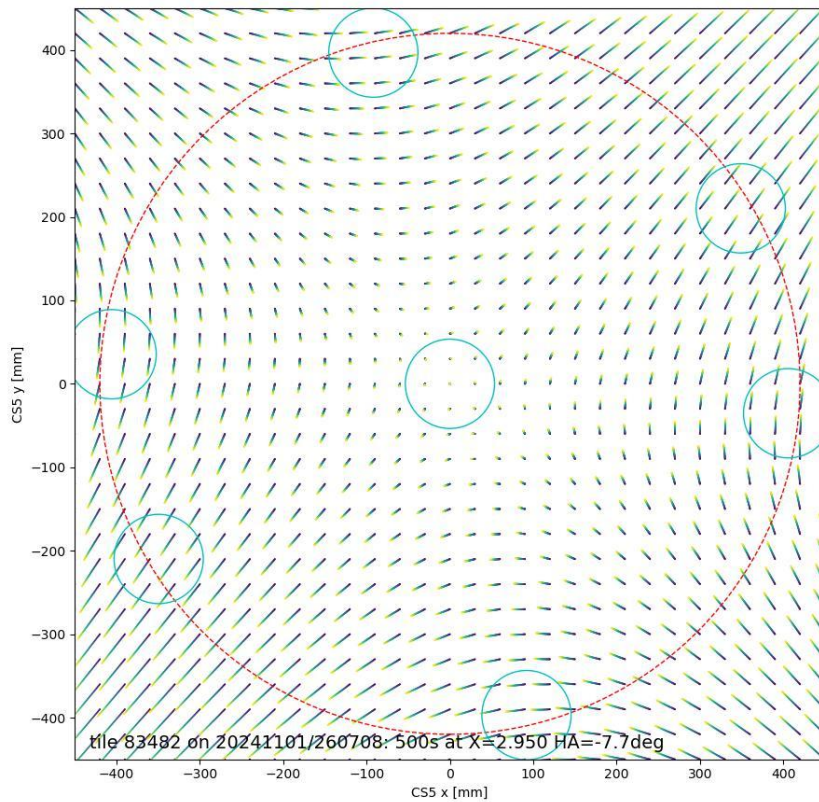
Guide Star Offsets in EXPID 260708 at X=3.0

Actual exposure time 1802s.



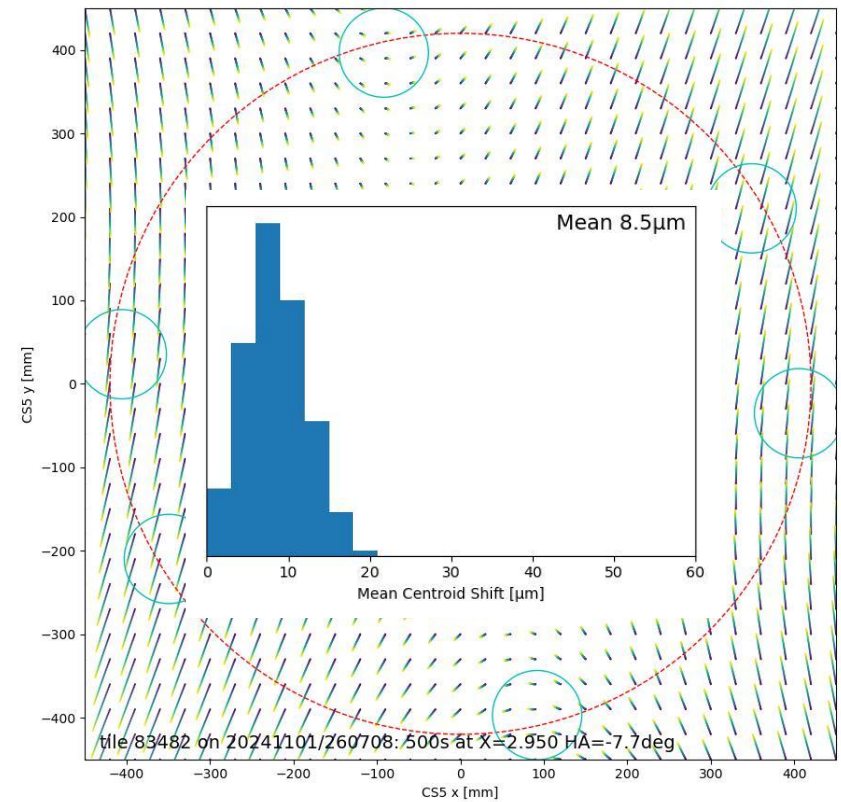
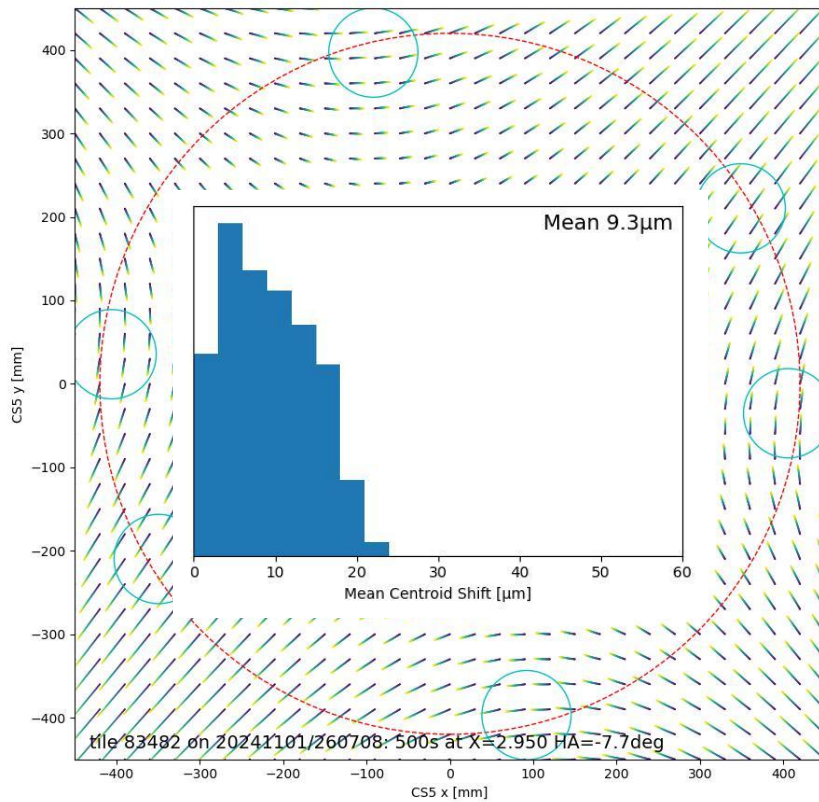
Guiding on Rotation?

We could use the centroid offsets provided by the guider to update the hexapod rotation during an exposure. However, rotating a quadrupole is $\sim$ zero sum:



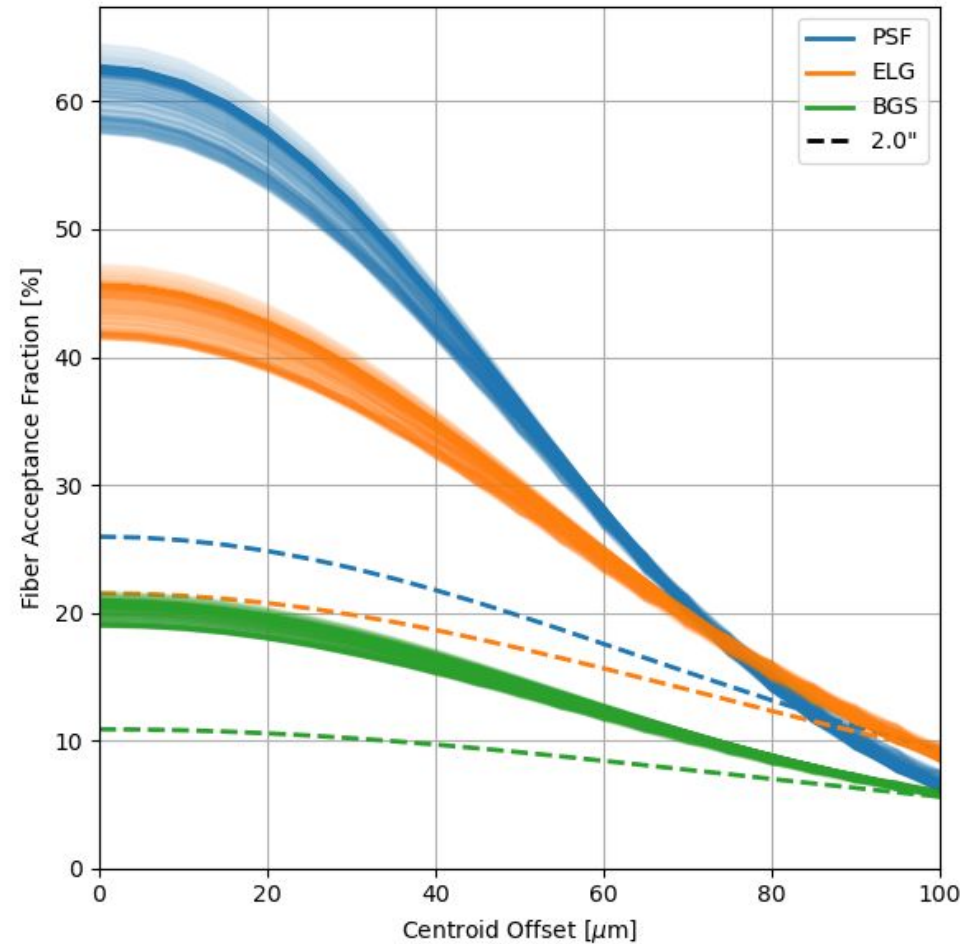
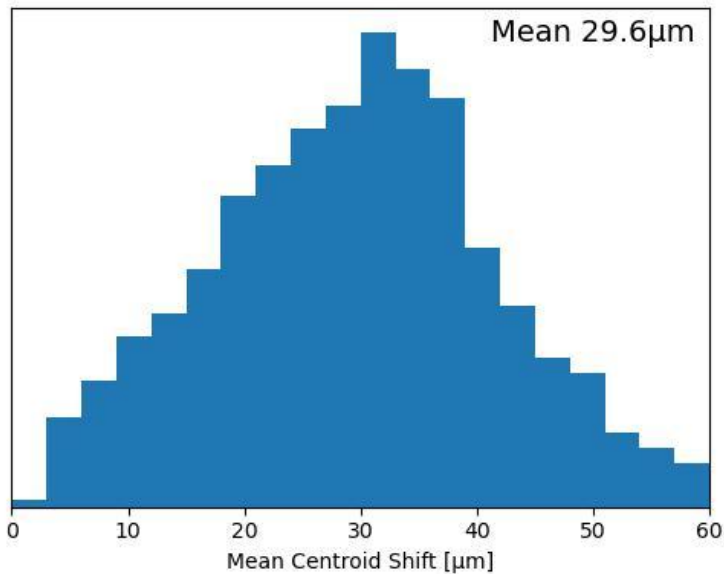
Guiding on Rotation?

We could use the centroid offsets provided by the guider to update the hexapod rotation during an exposure. However, rotating a quadrupole is $\sim$ zero sum:



Shorter Exposures

Some $X > 2$ exposures will probably need to be split early, depending on SBPROF (PSF/ELG/BGS) and seeing.



Summary

Ideal guiding performance is straightforward to calculate for a specified tile and observing conditions.

- expect large scale quadrupole pattern in throughput

Actual guiding performance matches ideal performance reasonably well.

- no guider improvements needed?

Shorter exposures will be needed for some tiles observed at $X > 2$

- depends on design DEC, initial HA & seeing
- decision can be made by NTS or integrated into ETC

